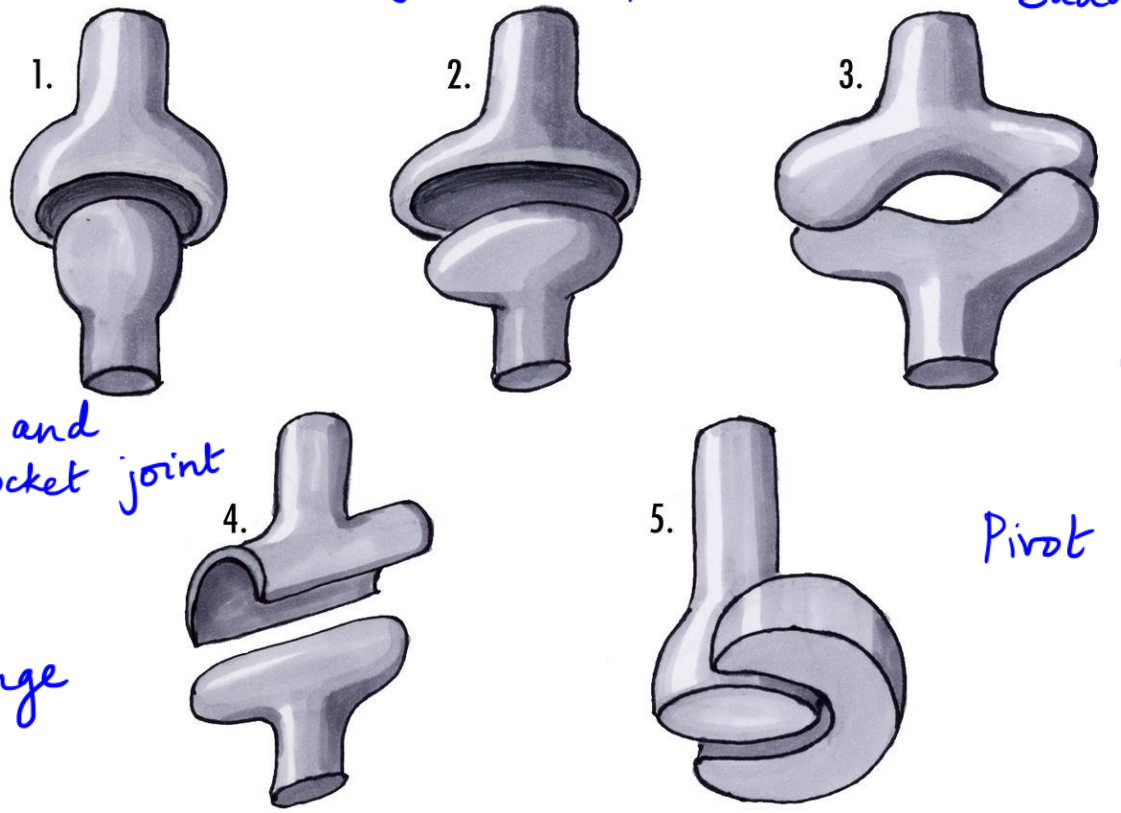
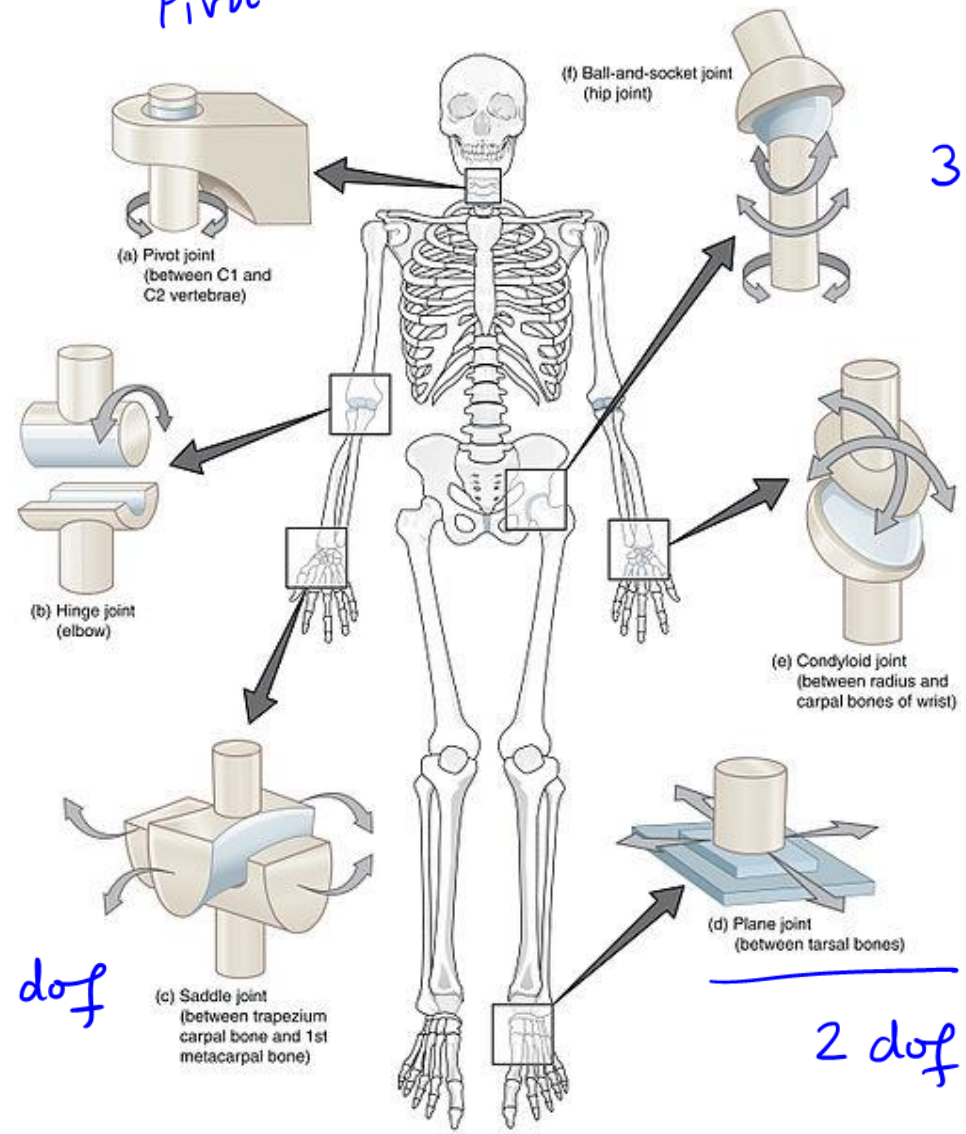
Types of synovial joints

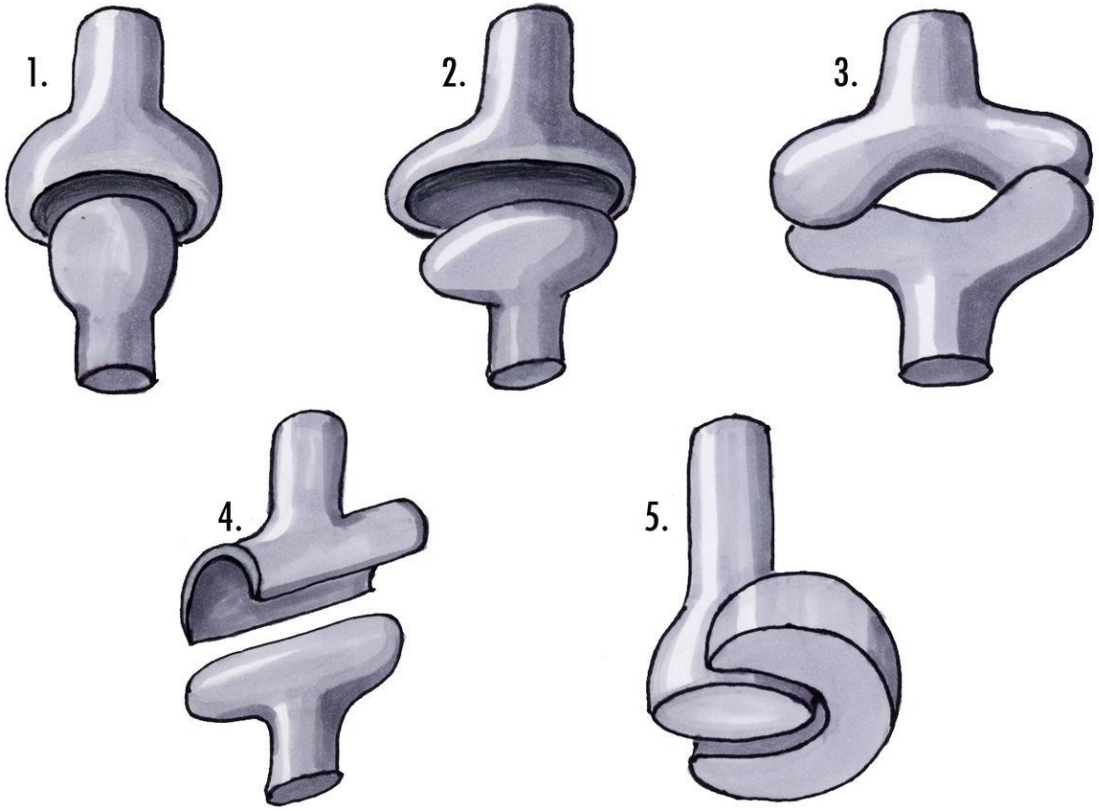


By Produnis - http://www.pflegewiki.de/index.php/Bild:Gelenke_Zeichnung01.jpg, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=337438

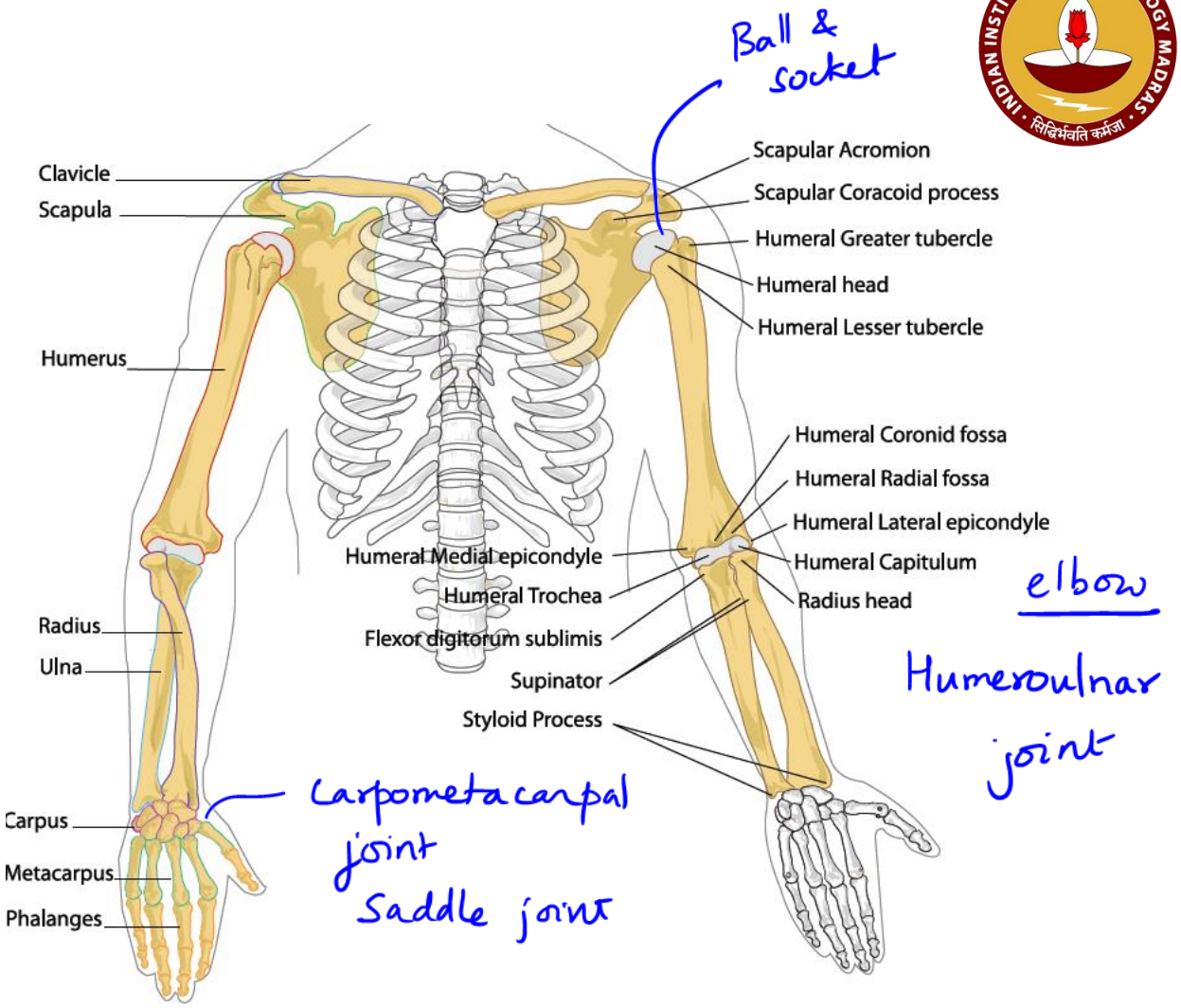


By OpenStax College [CC BY 3.0 (http://creativecommons.org/licenses/by/3.0)], via Wikimedia Commons

Types of synovial joints



By Produnis - http://www.pflegewiki.de/index.php/Bild:Gelenke_Zeichnung01.jpg, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=337438



By LadyofHats Mariana Ruiz Villarreal - Some of my sources: [1], Gray's Anatomy "Williams and Warwick", Atlas der anatomie des menschen "Sobotta",Anatomia del cuerpo humano "Yokochi, Rhoen, weinreb", and a lamina printed by "rdiger-anatomie GmbH"- LadyofHats, Public Domain, https://commons.wikimedia.org/w/index.php?curid=1522438

By OpenStax College [CC BY 3.0 (http://creativecommons.org/licenses/by/3.0)], via Wikimedia Commons

Types of synovial joints

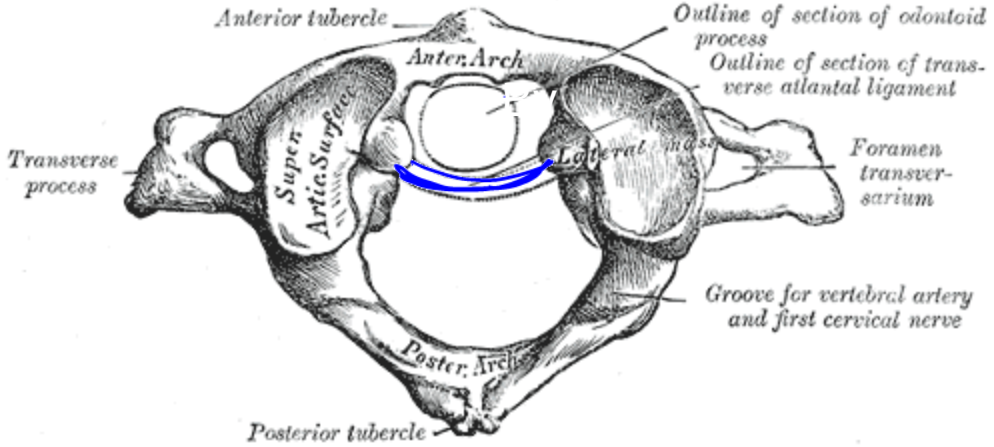
Atlas - C1

Axis - C2

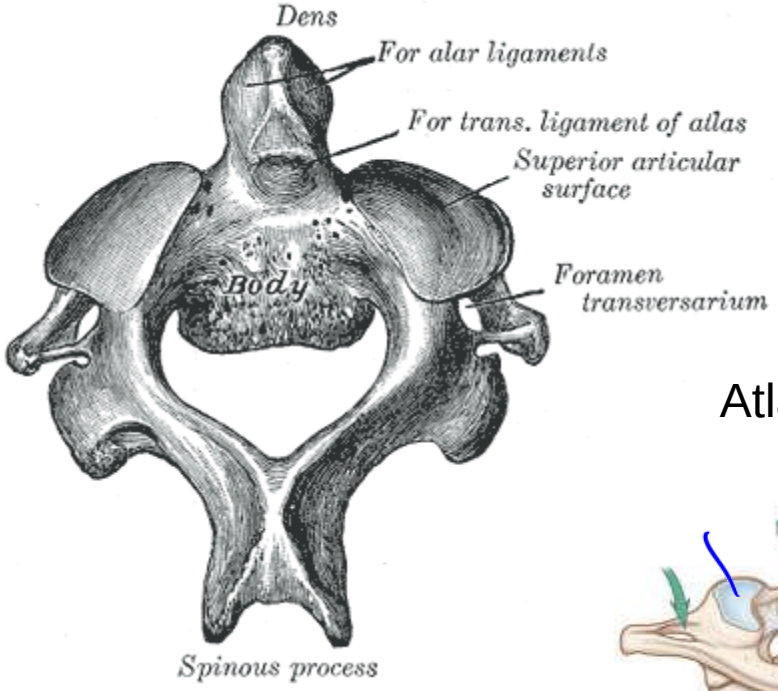
Pivot

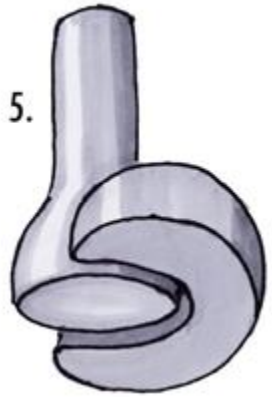
Axis

Atlas

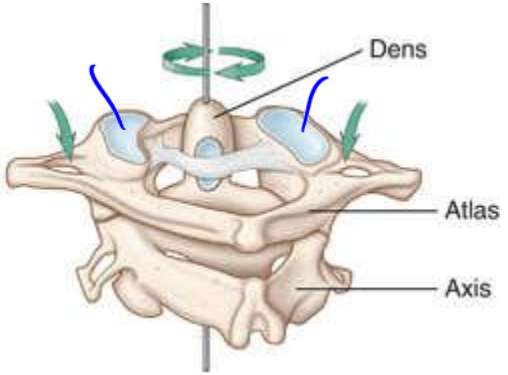


Henry Vandyke Carter [Public domain], via Wikimedia Commons



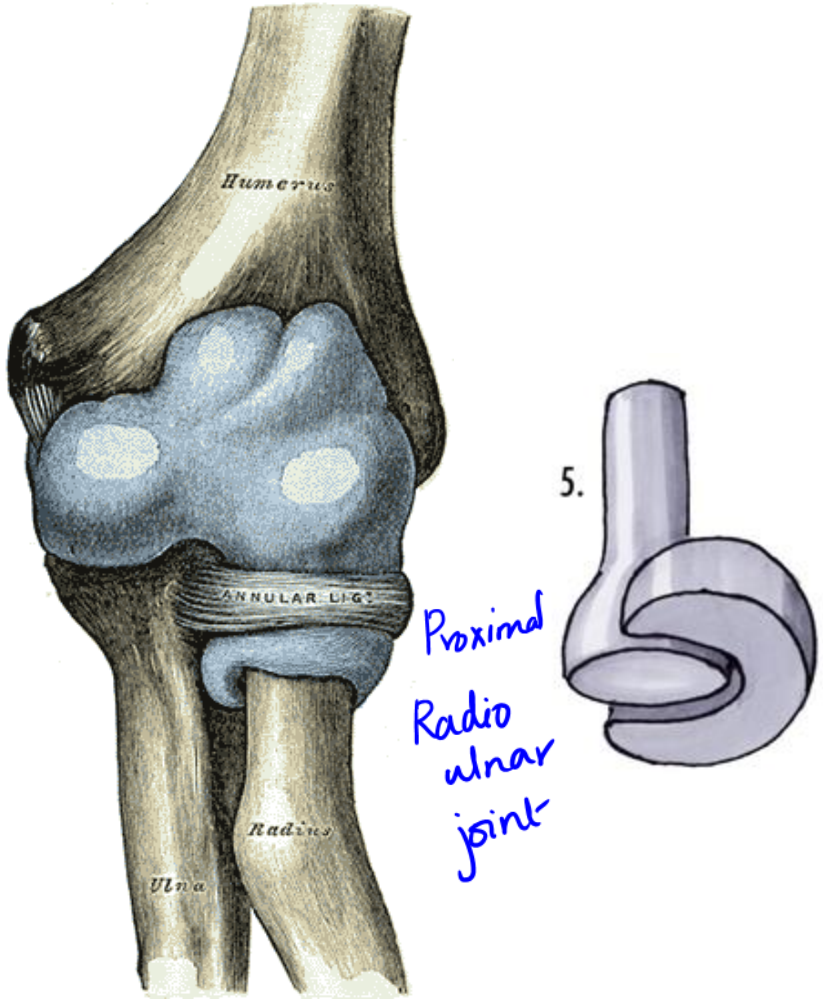


Atlantoaxial joint

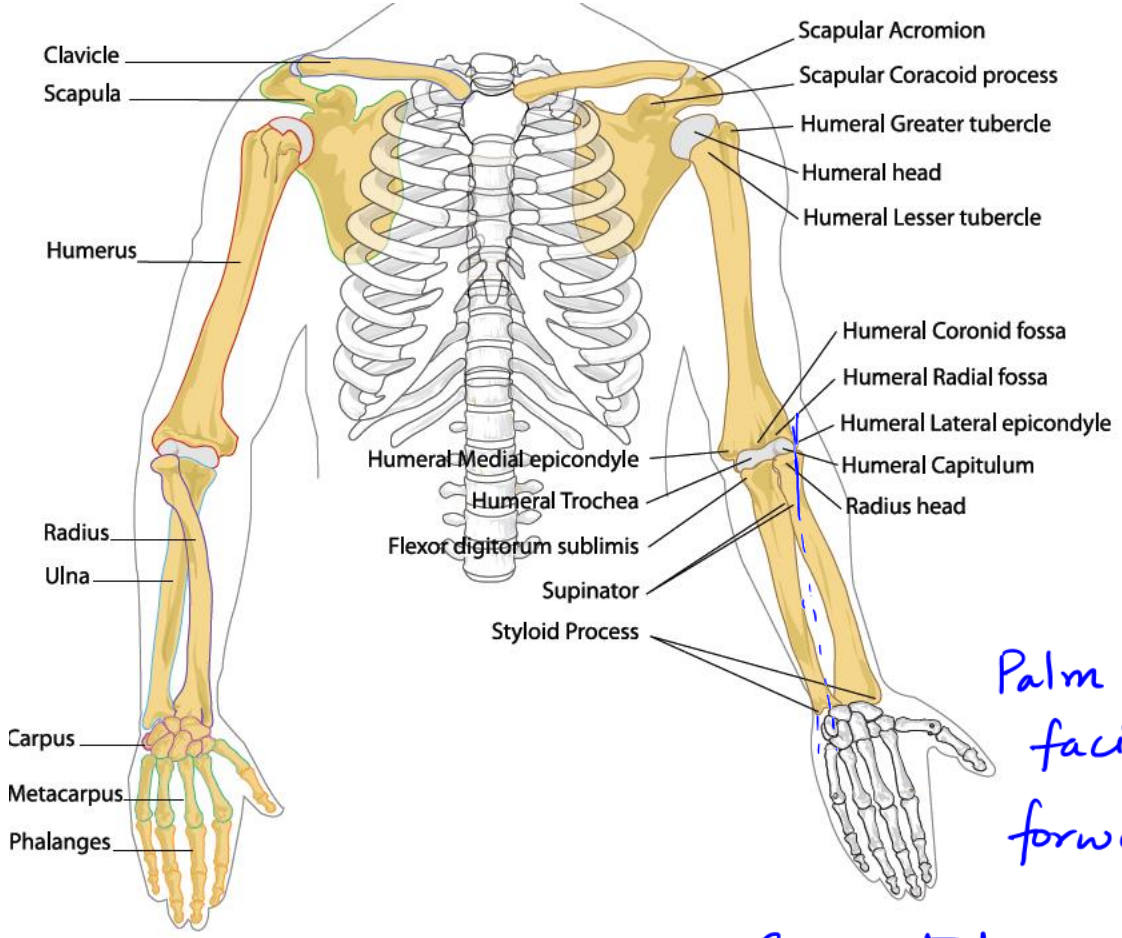


Henry Vandyke Carter - Henry Gray (1918) Anatomy of the Human Body (See "Book" section below)Bartleby.com: Gray's Anatomy, Plate 86, Public Domain, https://commons.wikimedia.org/w/index.php?curid=792296

Types of synovial joints



Proximal Radio ulnar joint



Pronated forearm

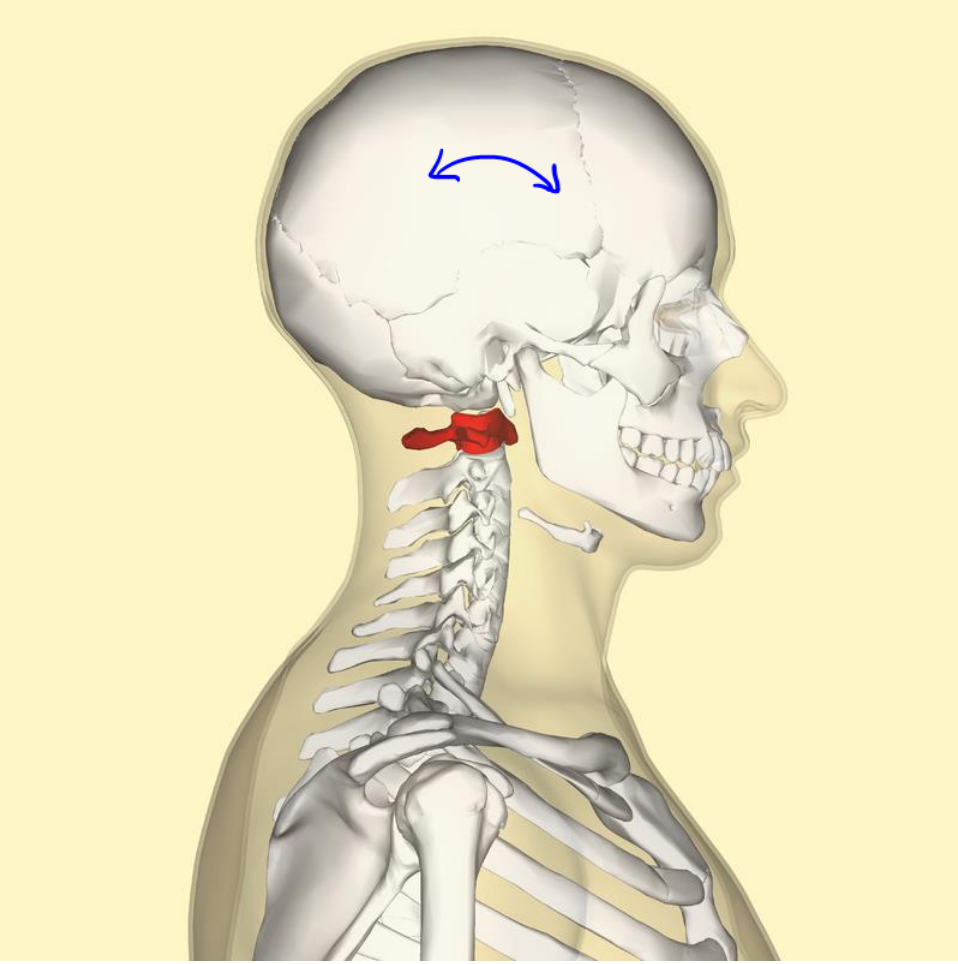
Palm facing forward

Supinated forearm

Henry Vandyke Carter [Public domain], via Wikimedia Commons

Types of synovial joints

Atlas C1 & the occipital bone



By Anatomography - en:Anatomography (setting page of this image), CC BY-SA 2.1 jp, https://commons.wikimedia.org/w/index.php?curid=22403918

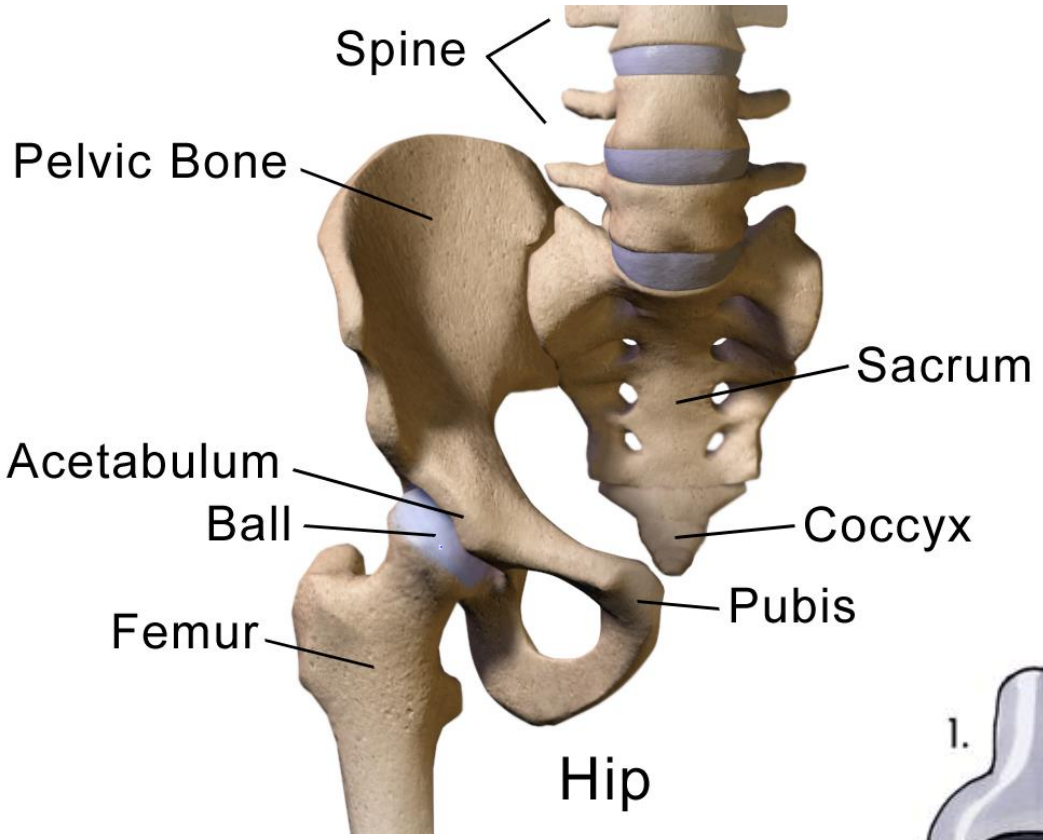
Atlantooccipital joint

condyloid joint

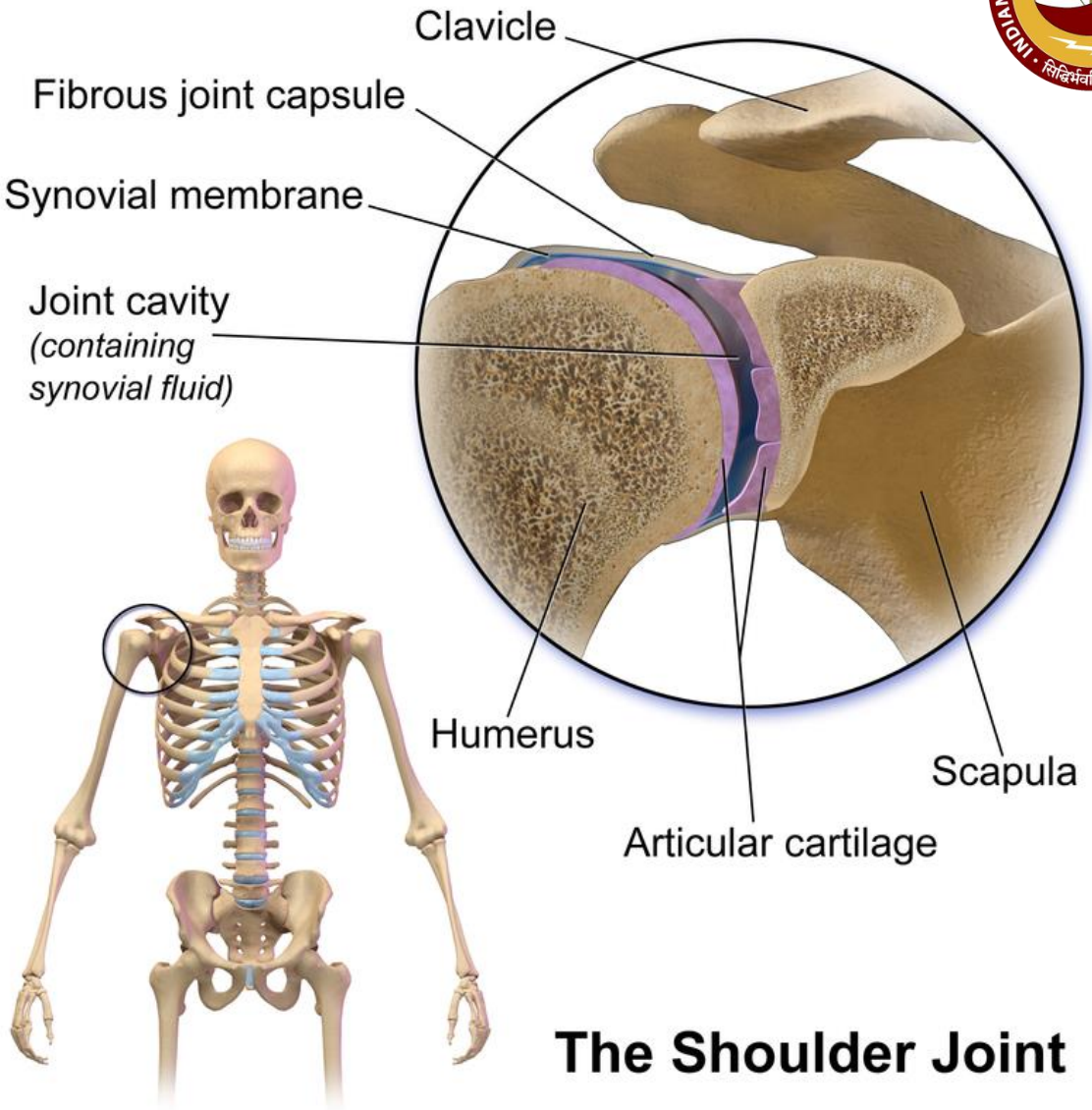


Types of synovial joints

By BruceBlaus - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=29987033



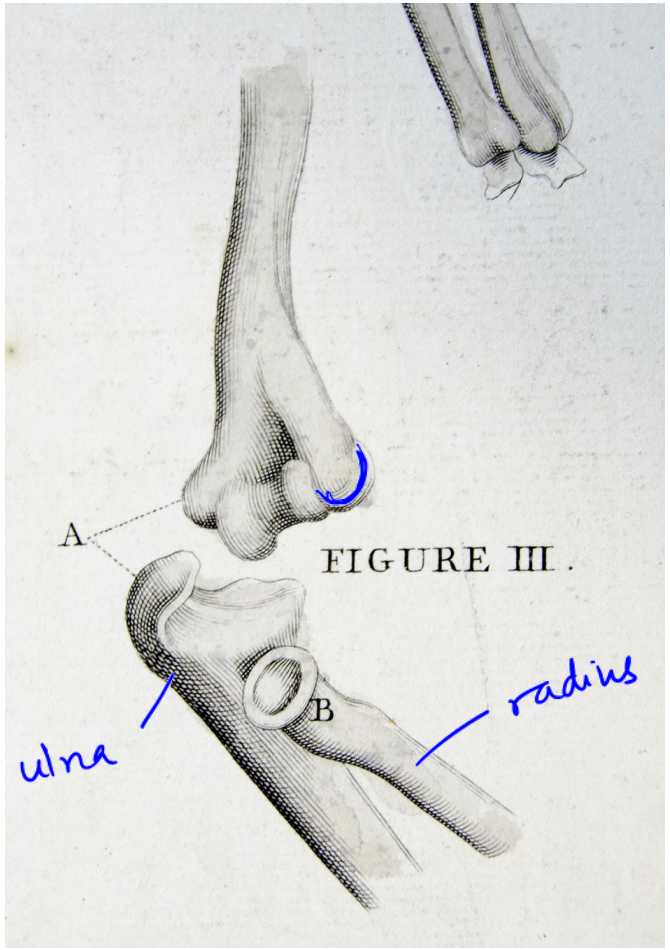




TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=27796970

Types of synovial joints



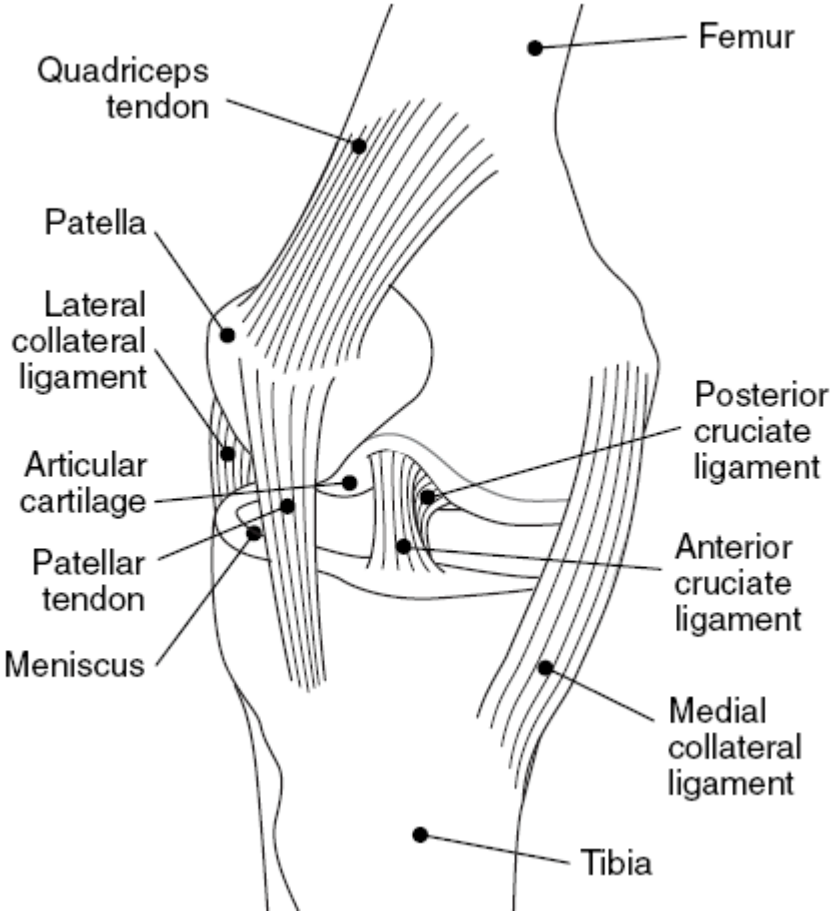
By University of Liverpool Faculty of Health from Liverpool, United Kingdom (elbow Uploaded by AlbertHerring) [CC BY 2.0 (http://creativecommons.org/licenses/by/2.0)], via Wikimedia Commons

Humeroulnar joint

Humeroradial joint

Radioulnar joint



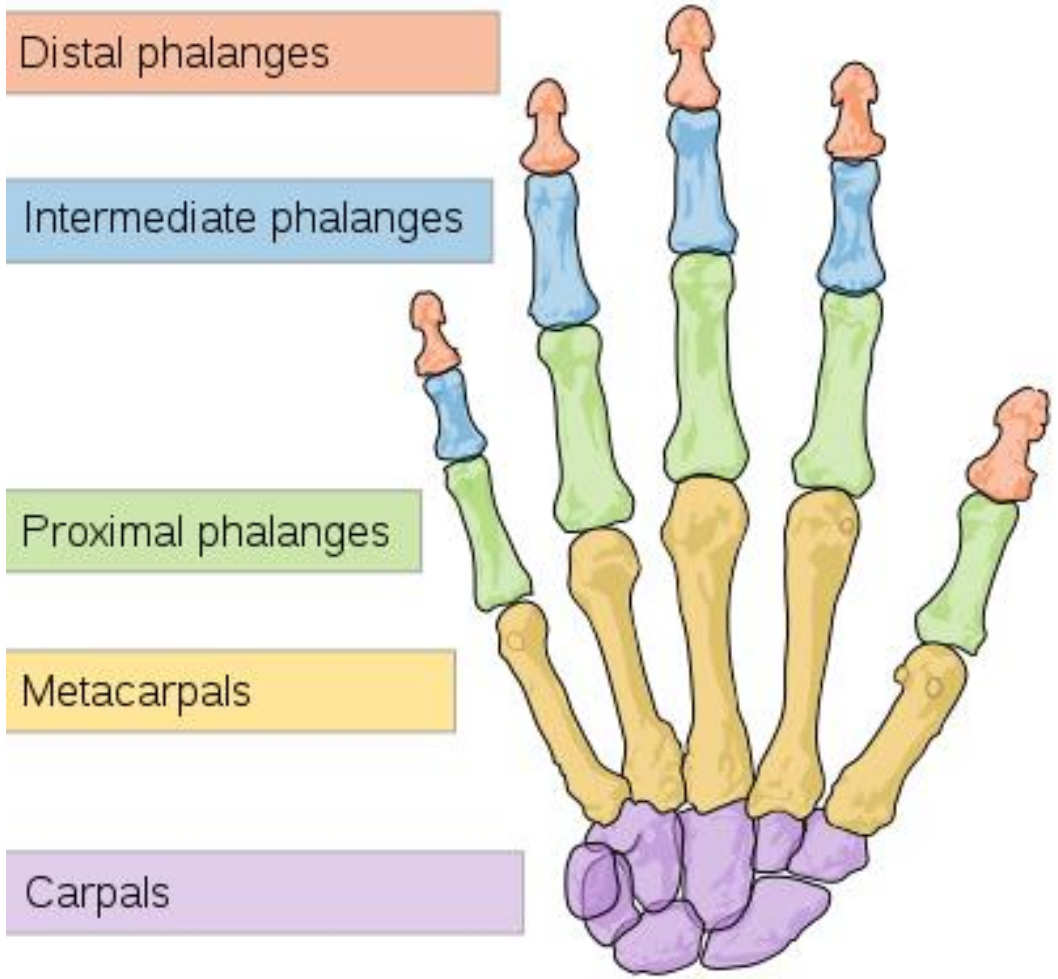


By National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS) - http://www.pueblo.gsa.gov/cic_text/health/qa-knee/kneeqa.htm, Public Domain, https://commons.wikimedia.org/w/index.php?curid=4957533

Types of synovial joints

Interphalangeal joints

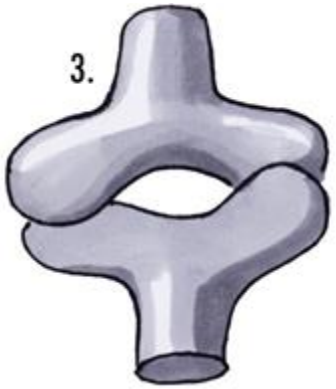


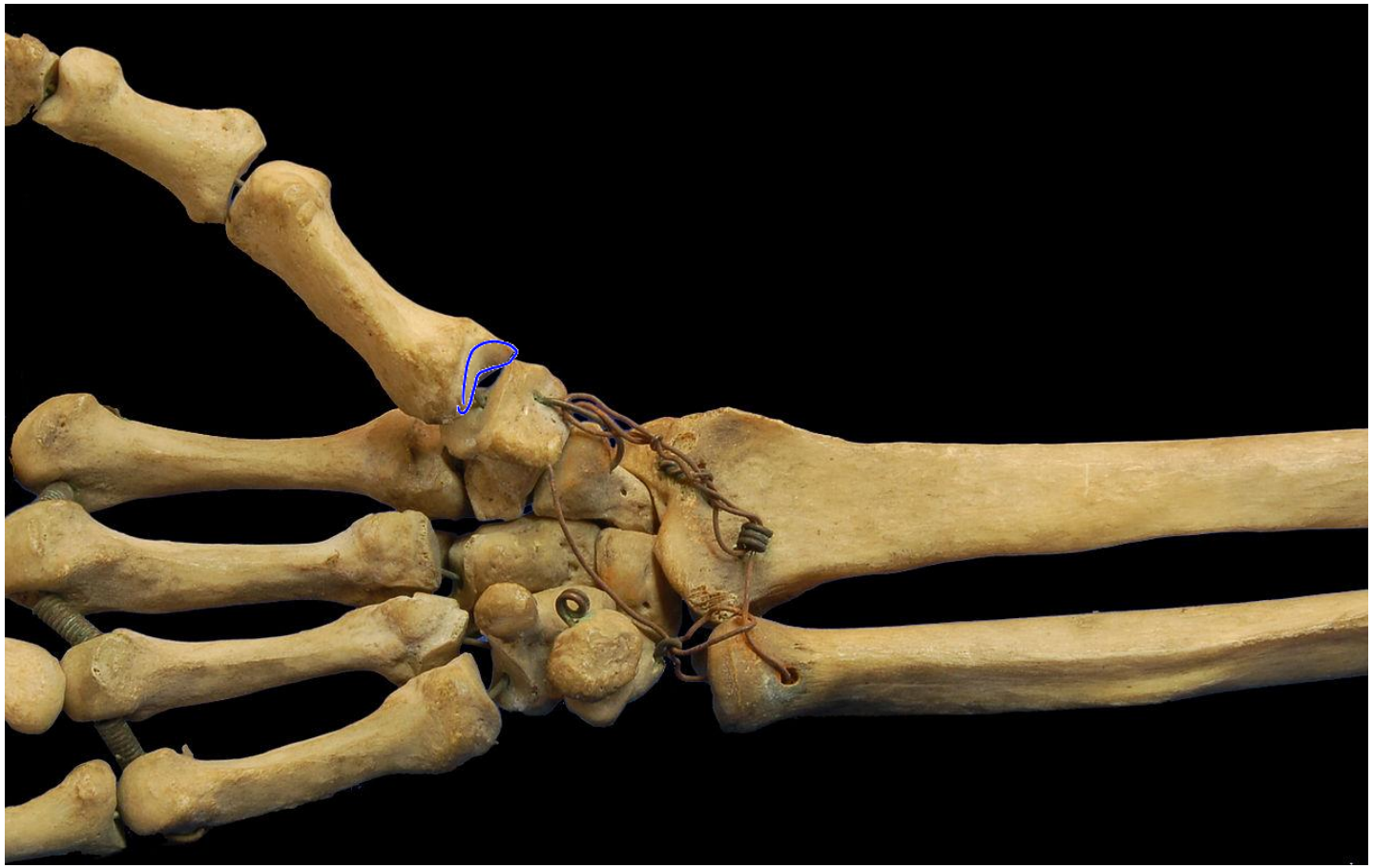


By Mariana Ruiz Villarreal (LadyofHats); retouches by Nyks - Own work. Image renamed from Image:Human hand bones simple-edit1-2.svg, Public Domain, https://commons.wikimedia.org/w/index.php?curid=3949051

Types of synovial joints

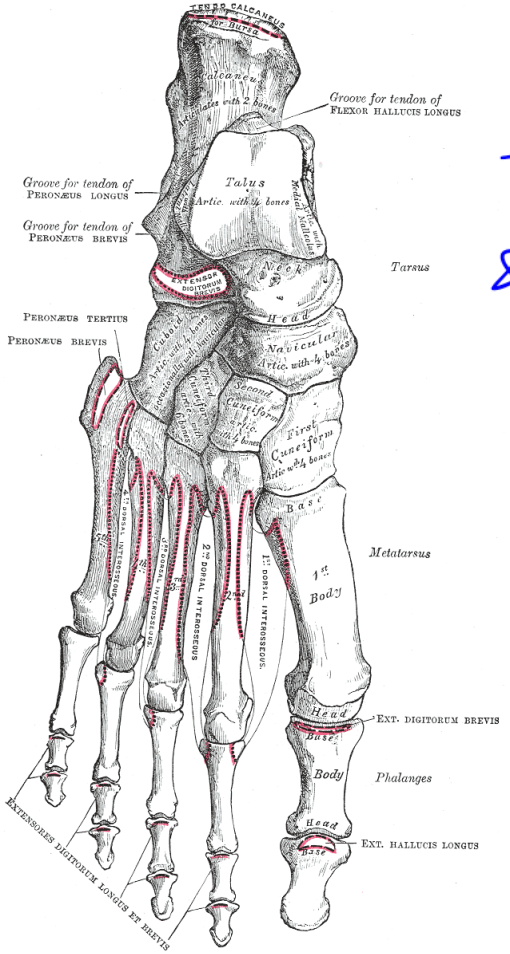
Carpometacarpal joint of the thumb
- Saddle joint





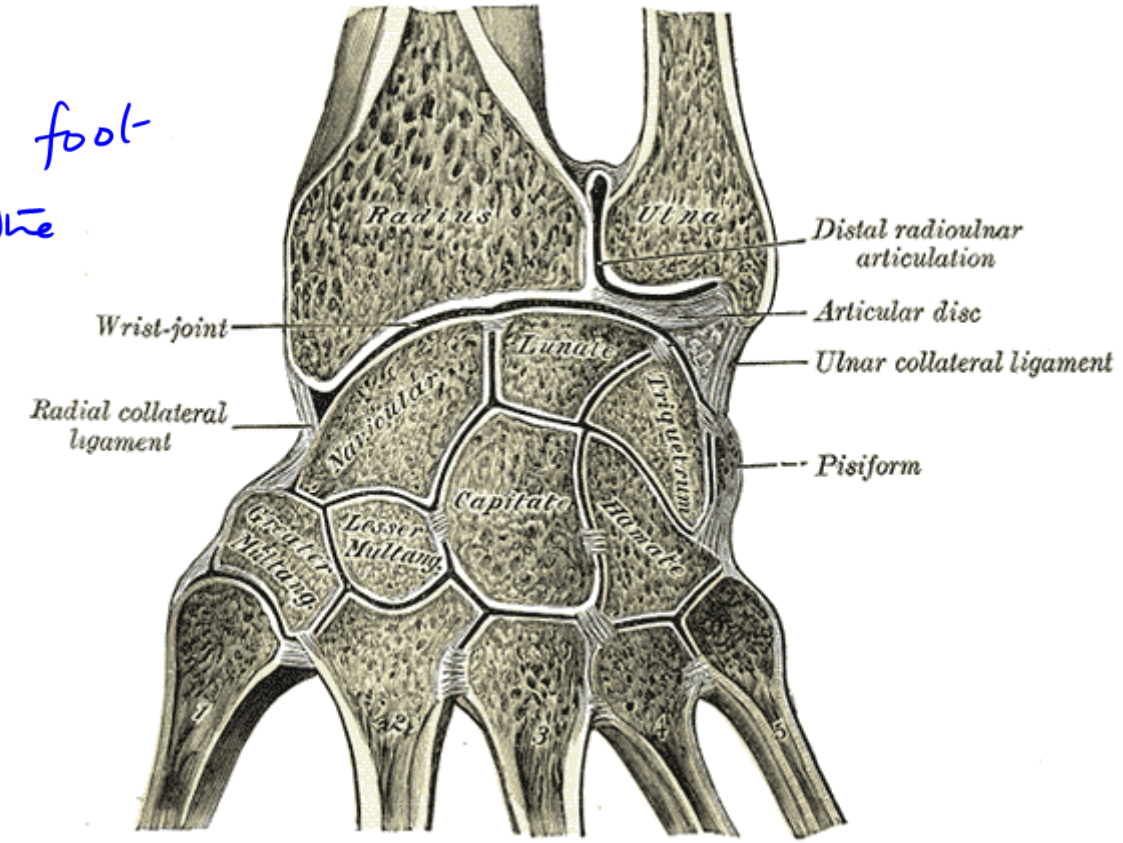
By Brian C. Goss (Own work) [Public domain], via Wikimedia Commons

Types of synovial joints



Planar

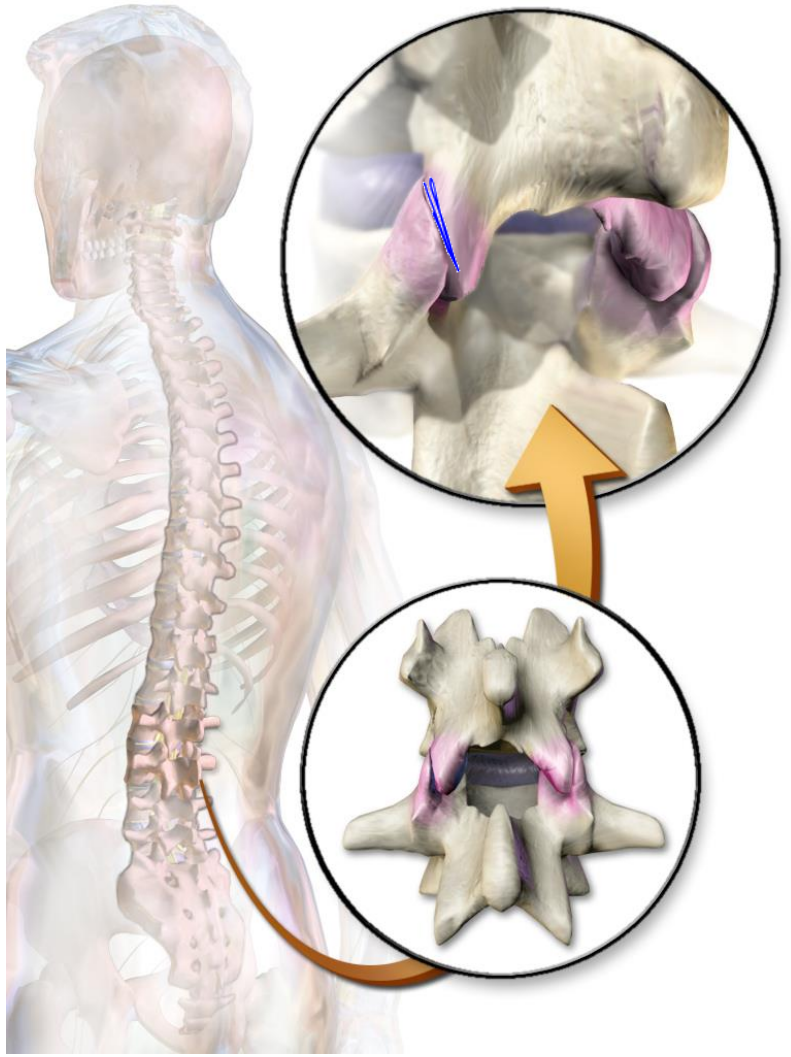
Tarsals in the foot & Carpals in the wrist



Henry Vandyke Carter [Public domain], via Wikimedia Commons

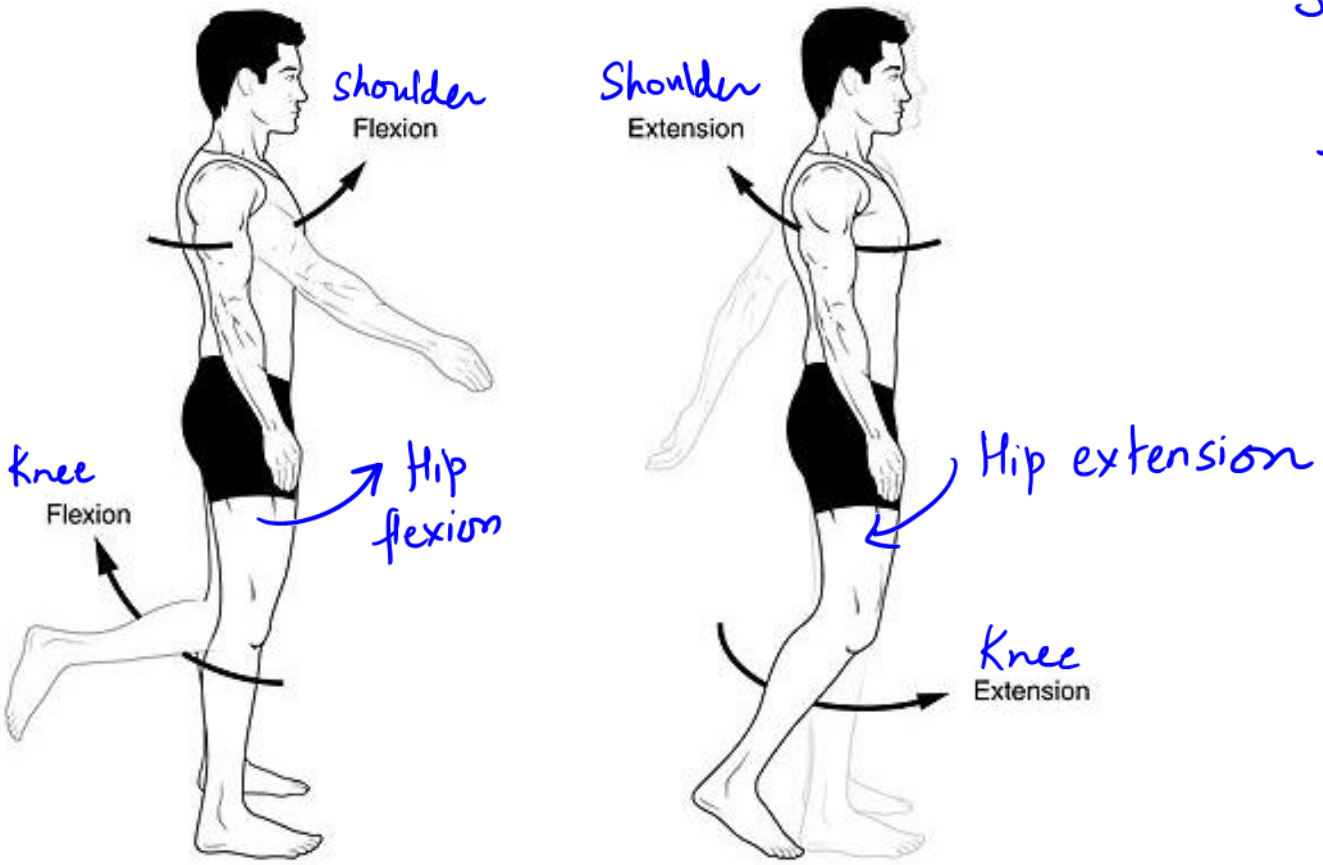
Types of synovial joints

Facet joints between two vertebrae

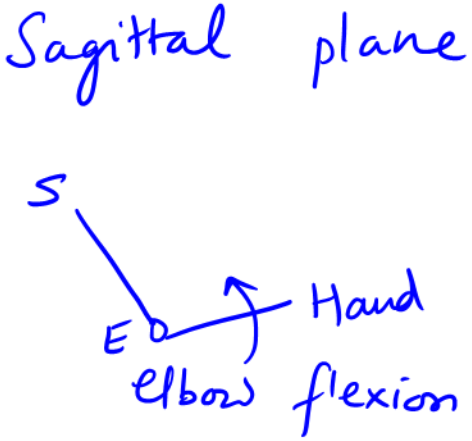


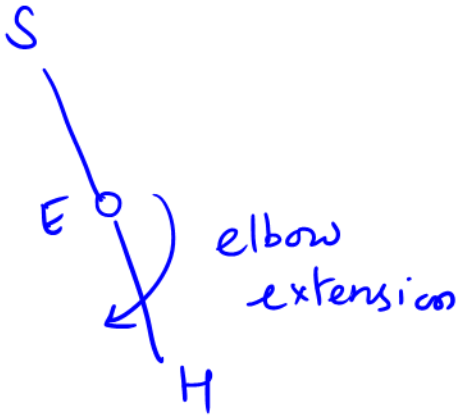
Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436. - Own work, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=27796926

Movements about joints



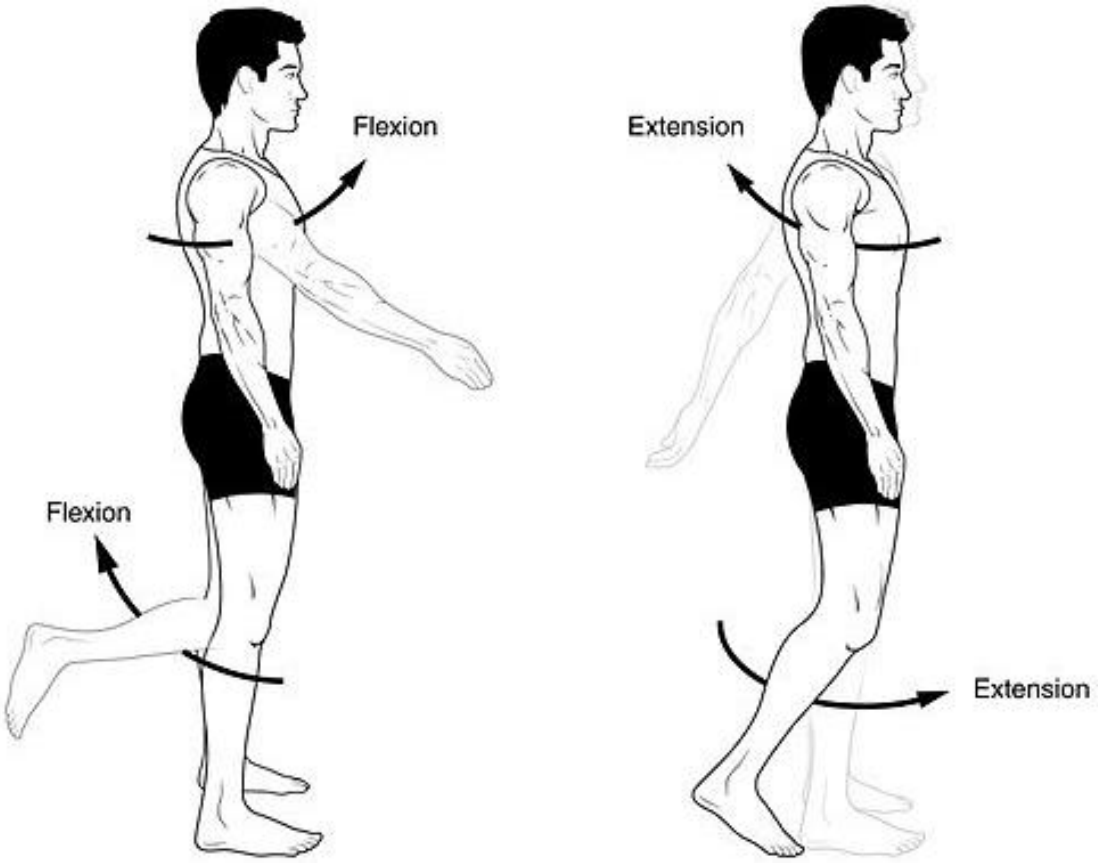
(a) and (b) Angular movements: flexion and extension at the shoulder and knees





By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

Movements about joints

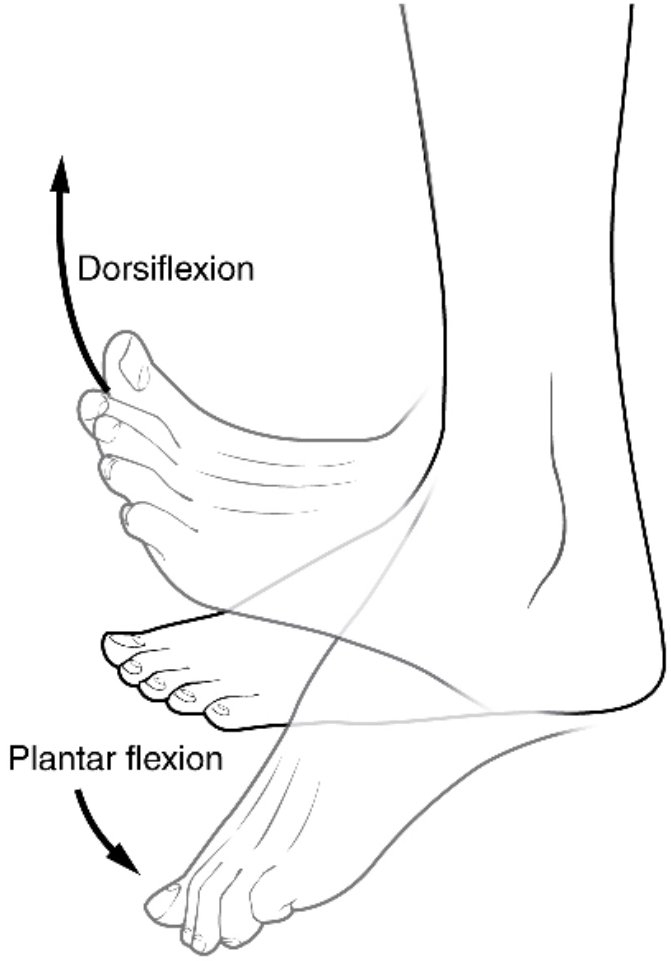


(a) and (b) Angular movements: flexion and extension at the shoulder and knees

Flexion - extn
occur about M-L axis

By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

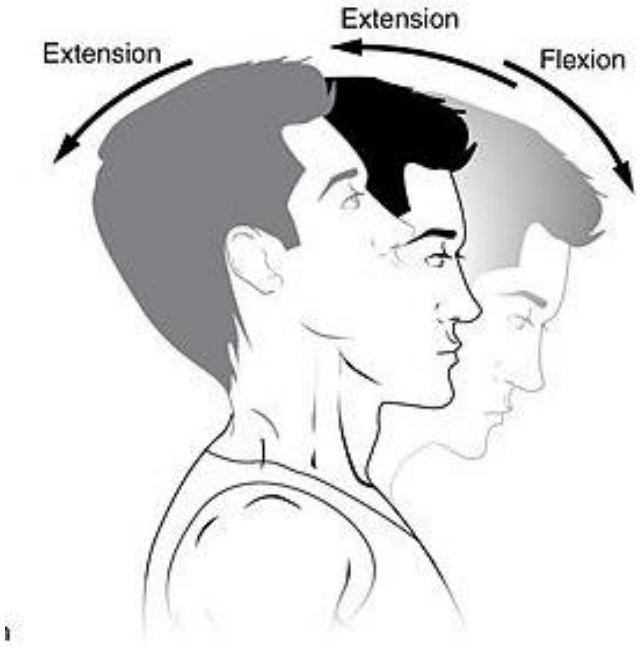
Movements about joints



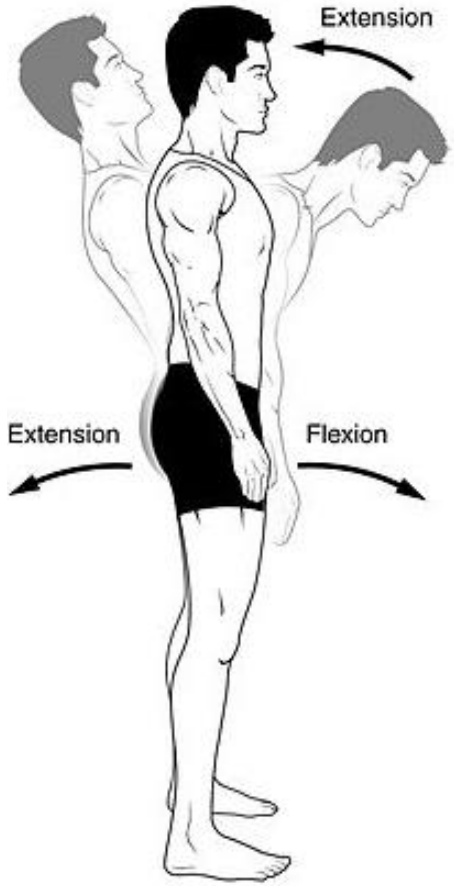
(h) Dorsiflexion and plantar flexion

By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

Movements about joints



(c) Angular movements: flexion and extension of the neck



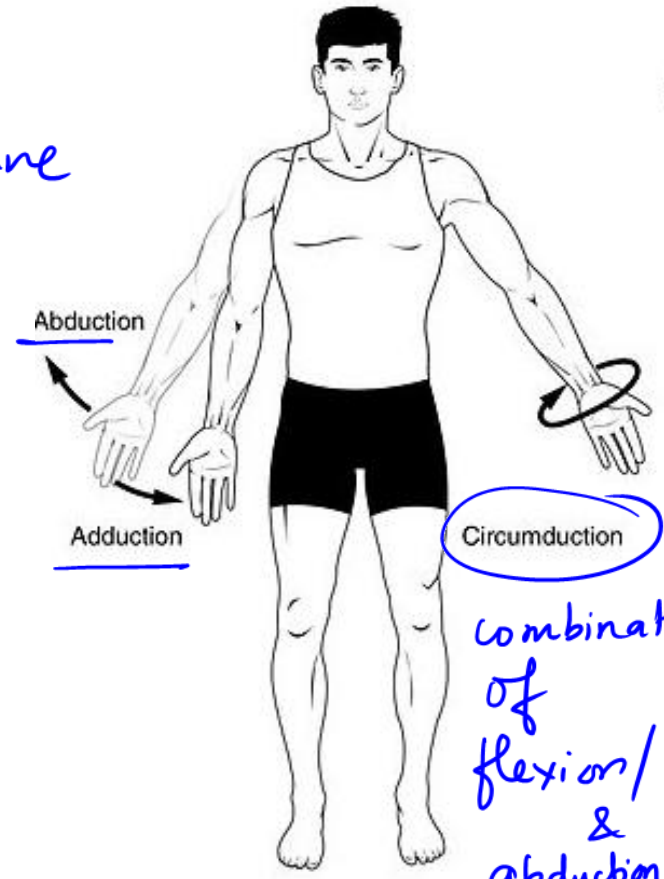
(d) Angular movements: flexion and extension of the vertebral column

By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

Movements about joints

Abduction - Adduction occur about A-P axis

Frontal plane



(e) Angular movements: abduction, adduction, and circumduction of the upper limb at the shoulder

combination of flexion/extn & abduction/adduction



(f) Rotation of the head, neck, and lower limb

Transverse plane

/ outward / external

/ inward / internal

Rotation about longitudinal axis or superior-inferior axis

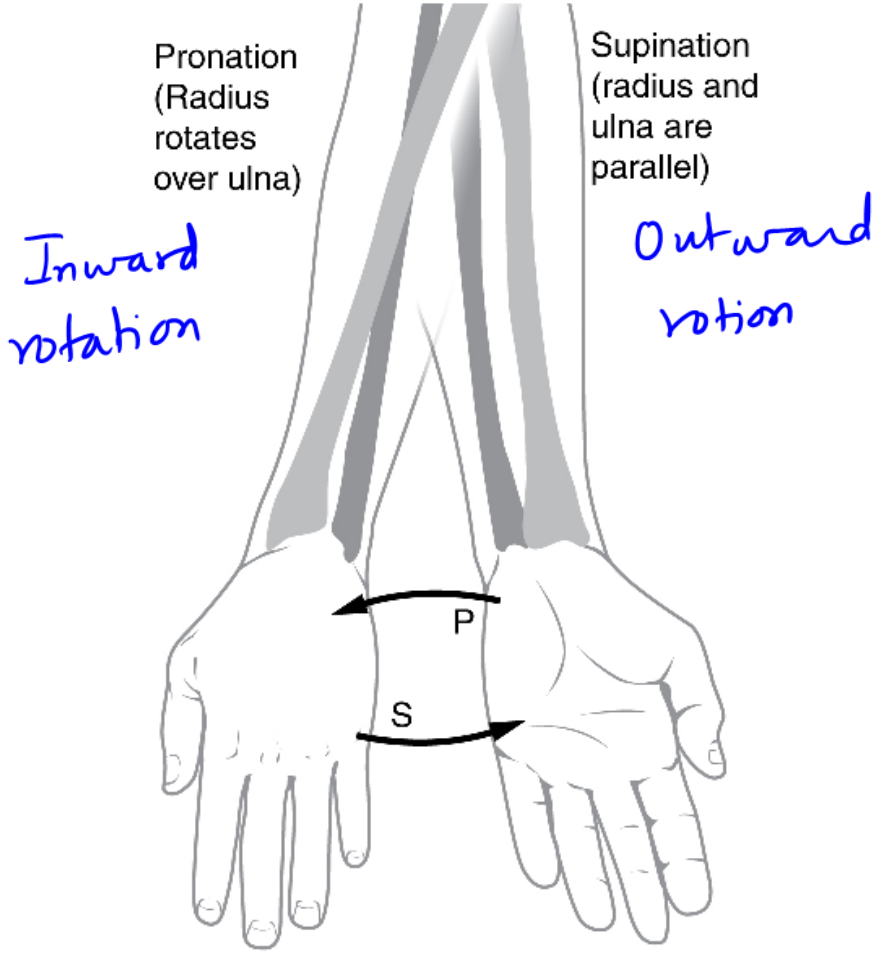
Wrist

Radial deviation

Ulnar deviation

By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

Movements about joints



(g) Pronation (P) and supination (S)



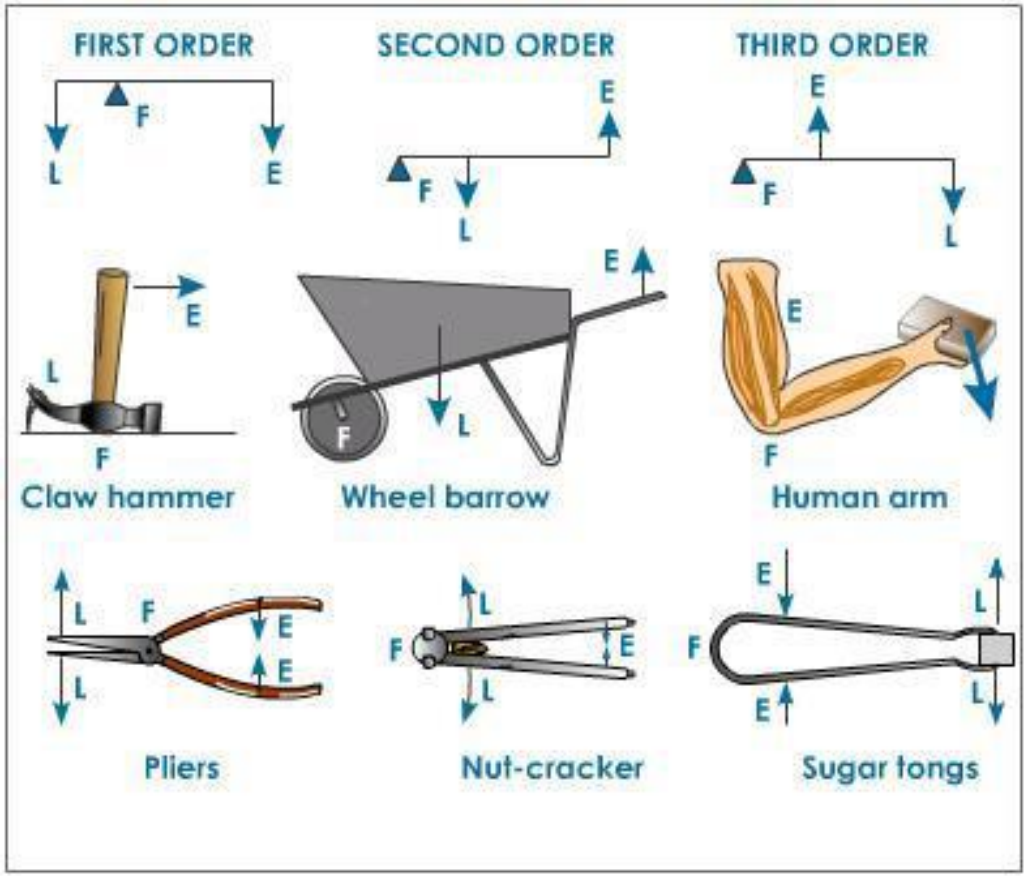
(i) Inversion and eversion

Frontal plane

By Tonye Ogele CNX [CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons

Levers in the human body

- Levers are simple machines, move a load (resistance) around a pivot (fulcrum) using a force (effort)
- Usually used to magnify force at the expense of distance
- Small effort moving a large load is called mechanical advantage



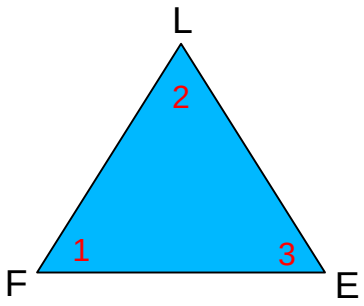
Source: https://alexeinstein.wordpress.com/2014/09/03/lever-of-human-body/. Used with permission

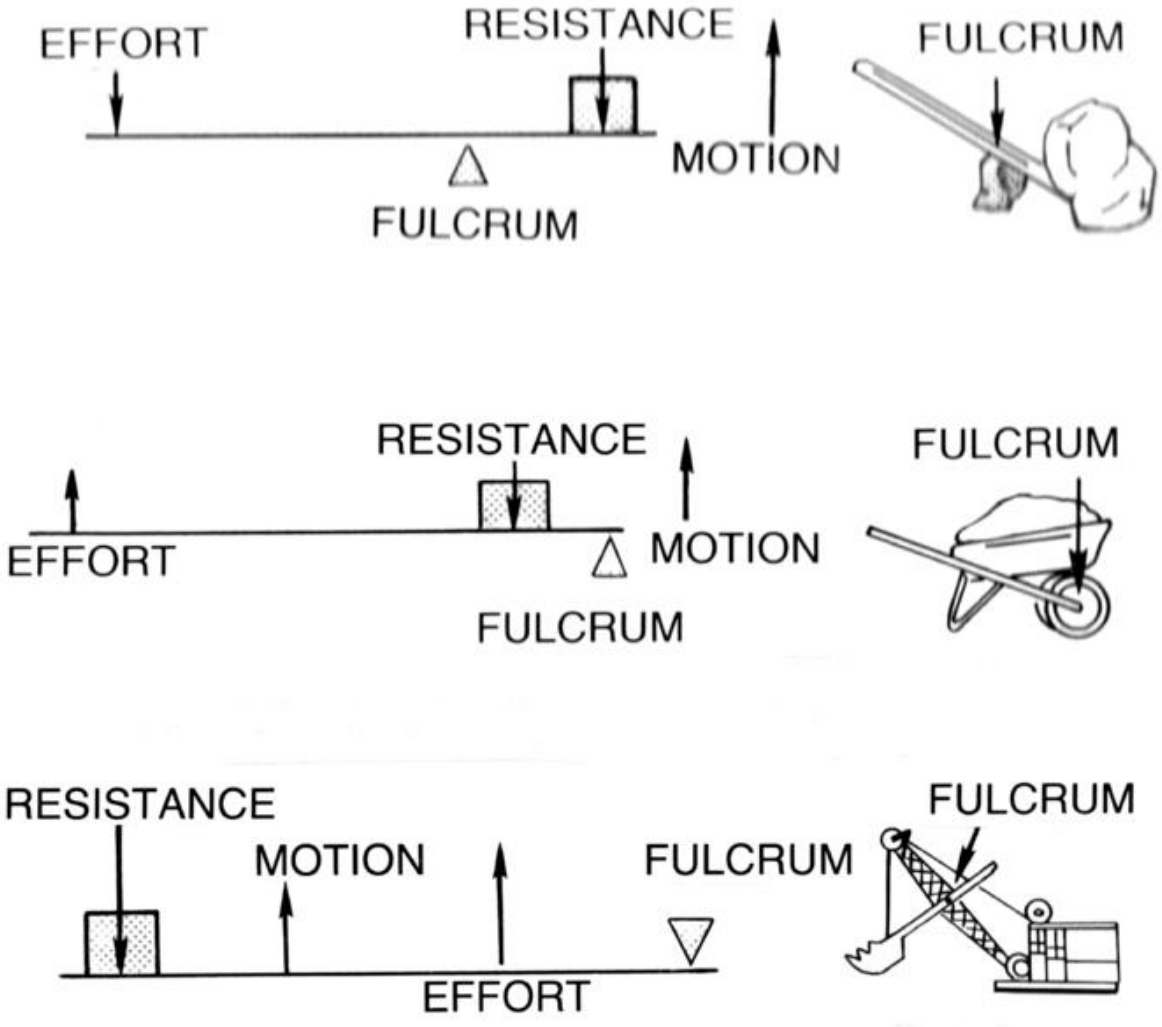
Classes of levers

First class
- Fulcrum between Load and Effort
- Load and Effort move in opposite directions

Second class
- Load between Effort and Fulcrum
- Load and Effort move in the same directions

Third class
- Effort between Fulcrum and Load
- Load and Effort move in the same directions

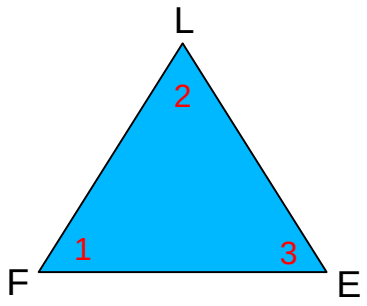




By Pearson Scott Foresman [Public domain], via Wikimedia Commons

Levers in the human body

- Levers of the musculoskeletal system tend to magnify motion at the expense of force
- Implies muscles produce much larger forces than those resisting the motion



Fulcrum – located at a joint

Effort – muscle force

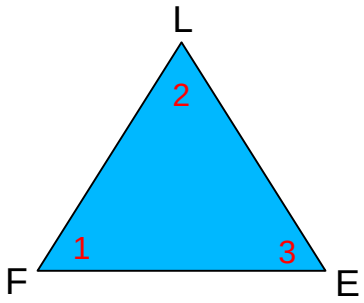
Load – external resistance

Levers in the human body

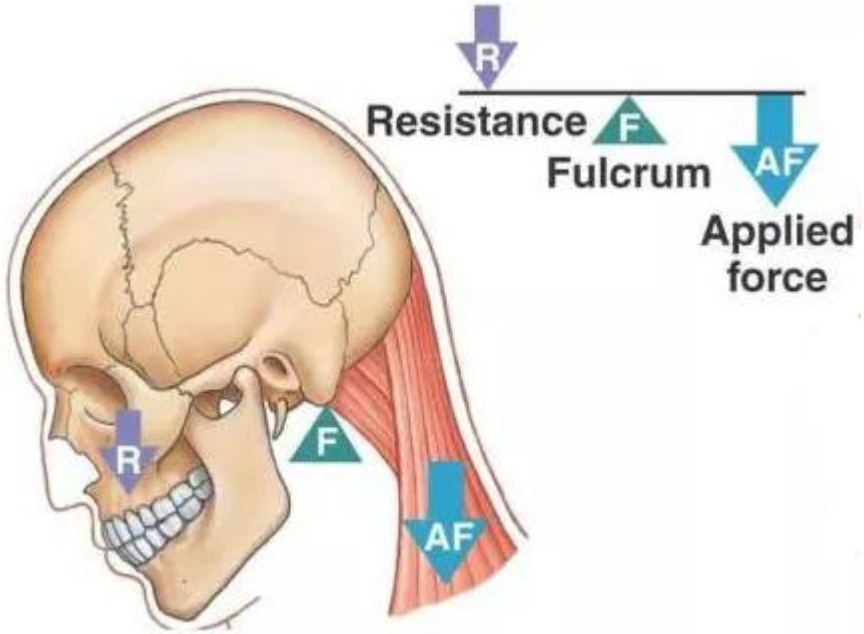
Fulcrum – located at a joint

Effort – muscle force

Load – external resistance



Lifting the head from a bent position



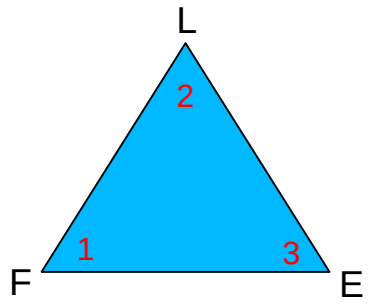
Source: https://alexeinstein.wordpress.com/2014/09/03/lever-of-human-body/. Used with permission

Levers in the human body

Fulcrum – located at a joint

Effort – muscle force

Load – external resistance



Going up on tiptoe



Source: https://alexeinstein.wordpress.com/2014/09/03/lever-of-human-body/. Used with permission

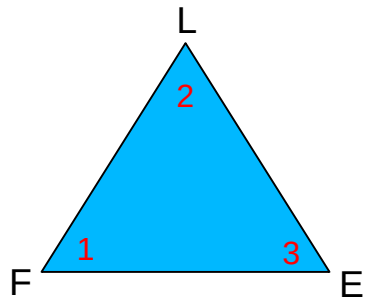
By w:en:Centers for Disease Control and Prevention - Growing Stronger - Strength Training for Older Adults - Exercises Stage 1, Public Domain, https://commons.wikimedia.org/w/index.php?curid=20629792

Levers in the human body

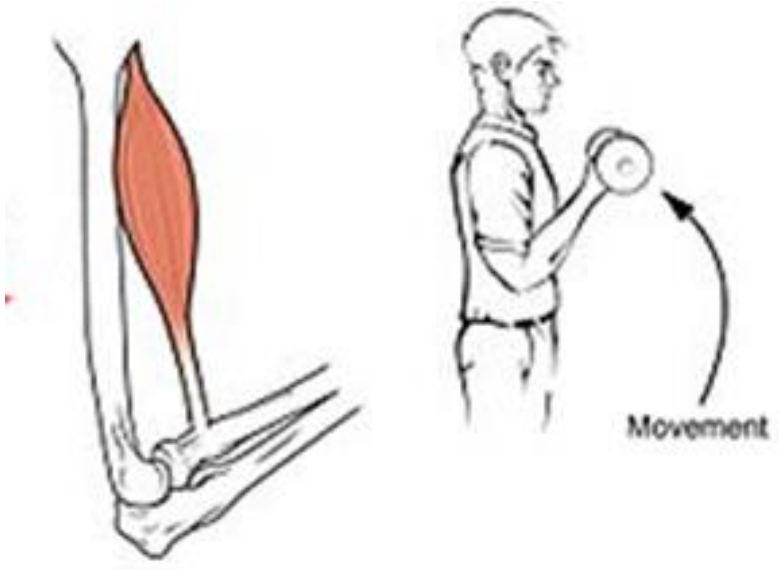
Fulcrum – located at a joint

Effort – muscle force

Load – external resistance



Lifting a load with the forearm



By OpenStax - https://cnx.org/contents/FPtK1zmh@8.25:fEI3C8Ot@10/Preface, CC BY 4.0, https://commons.wikimedia.org/w/index.php?curid=30015046 (cropped)

Levers in the human body

Fulcrum – located at a joint

Effort – muscle force

Load – external resistance

See

https://alexeinstein.wordpress.com/2014/09/03/lever-of-human-body/

